Visualization and Exploratory Analysis

AI-Assisted GitHub Monitoring Agent

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1 Introduction

To better understand the behavioral patterns captured by the monitoring agent, several visualizations were generated. These plots provide insight into how different features relate to student risk levels and help validate the effectiveness of the selected metrics.

2 Feature Correlation Analysis

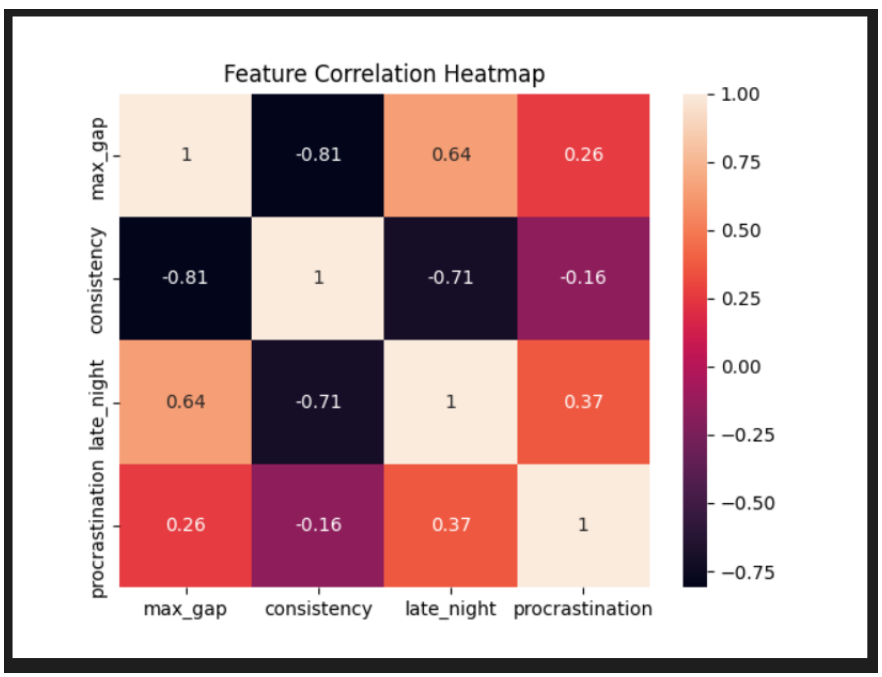

Figure 1: Correlation heatmap of key behavioral features

As shown in Figure 1, there is a strong negative correlation (-0.81) between *max gap* and *consistency*, indicating that students with long inactivity periods tend to exhibit irregular work patterns.

Additionally, *late-night activity* shows a moderate positive correlation (0.64) with *max gap*, suggesting that students who delay work are more likely to complete it during late hours. These relationships indicate that behavioral signals are interconnected and reinforce one another when identifying at-risk students.

3 Max Gap by Severity

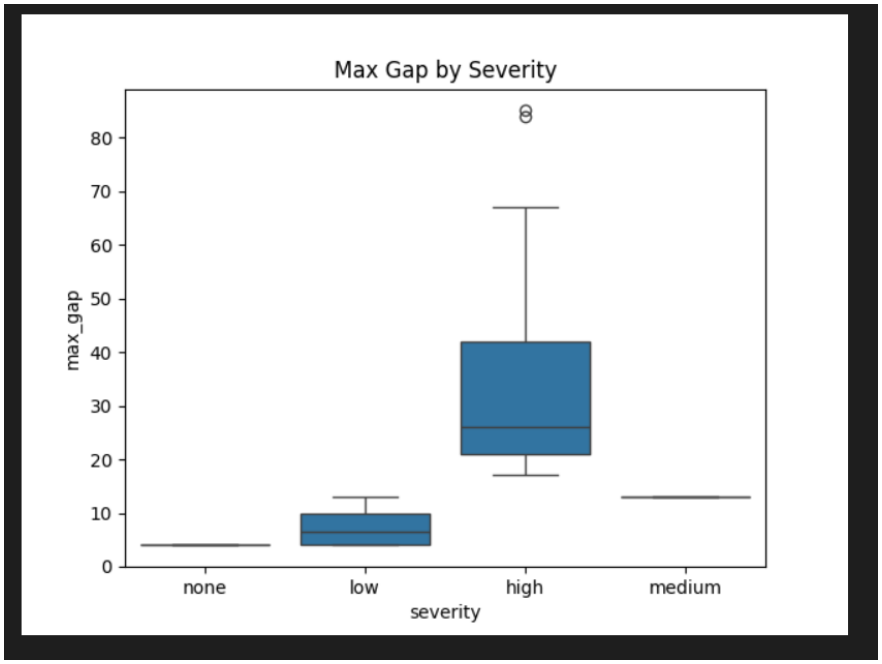

Figure 2: Distribution of maximum inactivity gap across severity levels

Figure 2 shows a clear separation between severity levels based on inactivity. Students in the *high severity* group exhibit significantly larger gaps between commits, with some extreme cases exceeding 80 days. In contrast, students in the *none* category maintain consistently low inactivity gaps.

This confirms that prolonged inactivity is one of the strongest indicators of struggling behavior.

4 Late-Night Work Patterns

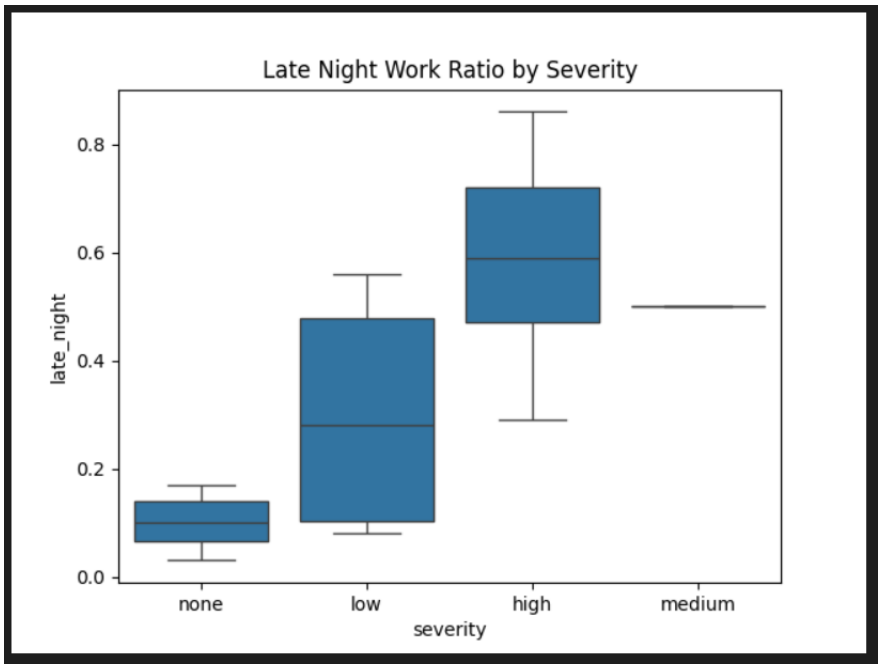

Figure 3: Late-night work ratio across severity levels

As shown in Figure 3, late-night activity increases with severity. Students with no identified risk typically have low late-night work ratios, while high-severity students frequently exceed 0.5.

This suggests that struggling students are more likely to adopt irregular work schedules, potentially due to procrastination or time pressure.

5 Additional Observations

Although not shown, the procrastination and severity distribution plots provide additional context:

- Procrastination shows overlap across severity levels, explaining its weaker predictive performance.
- The dataset exhibits class imbalance, with a higher number of high-severity students, which contributes to model bias during validation.

6 Key Insights

From these visualizations, several important conclusions can be drawn:

- **Max gap is the strongest predictor of student risk.**
- **Late-night activity provides supporting evidence of irregular work behavior.**
- **Behavioral features are correlated but capture different aspects of engagement.**
- **Class imbalance impacts model performance and prediction bias.**

7 Conclusion

These visualizations validate the design of the monitoring agent and demonstrate that behavioral patterns derived from GitHub activity can effectively identify at-risk students. The results also highlight areas for improvement, particularly in handling intermediate severity levels and addressing dataset imbalance.